

Madhava Nallamilli

CS 230 : Discrete Computational Structures

Summer Semester, 2025
HOMEWORK ASSIGNMENT #3
Due Date: Friday, June 13

Suggested Reading: Rosen Sections 2.1 - 2.3; Lehman et al. Chapter 4.1, 4.3, 4.4., Lecture Notes Functions , Relations; Rosen Sections 9.1 and 9.5; Lehman et al. Chapter 10.5, 10.6 and 10.10

For the problems below, explain your answers and show your reasoning.
The total points add up to 100, but you will be graded out of 90. This means you have the opportunity to earn 10 extra credit points.

Functions

1. **[10 pts]** Prove that $f(n) = 5n + 9$ is one-to-one, where the domain and co-domain of f is $\mathbb{Z}^+$. Show that f is not onto.

If $f(a) = f(b)$, then $a = b \Rightarrow f(n) = 5n + 9$ is one-to-one

$$f(a) = f(b)$$

$$\Rightarrow 5a + 9 = 5b + 9$$

$$\Rightarrow 5a = 5b$$

$$\Rightarrow a = b$$

Therefore, f is one-to-one.

For every $y \in \mathbb{Z}^+$, there must exist an $n \in \mathbb{Z}^+$ such that:

$$f(n) = y \Rightarrow y = 5n + 9 \Rightarrow n = \frac{y-9}{5}$$

Not all positive integers y make $\frac{y-9}{5}$ positive integer:

$$\text{If } y = 1, \text{ then } n = \frac{1-9}{5} = -\frac{8}{5} \notin \mathbb{Z}^+$$

$$\text{If } y = 10, \text{ then } n = \frac{10-9}{5} = \frac{1}{5} \notin \mathbb{Z}^+$$

So, there exist elements in $\mathbb{Z}^+$ that are not in the range of f .

Hence, f is not onto.

2. [10 Pts] Prove that $f(A, B) = A \cap B$ is onto, where the domain of f is $\mathbf{P}(\mathbf{N}) \times \mathbf{P}(\mathbf{N})$ and the co-domain of f is $\mathbf{P}(\mathbf{N})$. Show that f is not one-to-one.

Onto:

Let $C \in \mathbf{P}(\mathbf{N})$.

Define: $A = B = C$.

$$f(A, B) = A \cap B = C \cap C = C$$

Therefore, for every subset $C \subseteq \mathbf{N}$, there exists a pair $(A, B) \in \mathbf{P}(\mathbf{N}) \times \mathbf{P}(\mathbf{N})$ such that $f(A, B) = C$. Thus, f is onto.

One-To-One:

To show that f is not one-to-one, there has to be two different pairs $(A_1, B_1) \neq (A_2, B_2)$ such that:

$$A_1 \cap B_1 = A_2 \cap B_2$$

Consider the following two distinct input pairs:

$$A_1 = \{1, 2, 3\}, \quad B_1 = \{2, 3, 4\}$$

$$A_2 = \{2, 3\}, \quad B_2 = \{2, 3\}$$

Then,

$$f(A_1, B_1) = A_1 \cap B_1 = \{2, 3\}$$

$$f(A_2, B_2) = A_2 \cap B_2 = \{2, 3\}$$

$(A_1, B_1) \neq (A_2, B_2)$, so f is not one-to-one.

3. [8 Pts] Let $f(A) = A \cup \{1\}$, where the domain and co-domain of f is $\mathbf{P}(\mathbf{N})$. Prove that f is neither one-to-one nor onto.

Not One-To-One:

To prove that f is not one-to-one, I have to find two distinct sets $A, B \in \mathbf{P}(\mathbf{N})$ such that

$$A \neq B \text{ but } f(A) = f(B).$$

Let $A = \emptyset$ and $B = \{1\}$. Then:

$$f(A) = \emptyset \cup \{1\} = \{1\}, \quad f(B) = \{1\} \cup \{1\} = \{1\}.$$

$A \neq B$ but $f(A) = f(B)$, so f is not one-to-one.

Not Onto:

To prove that f is not onto, I have to find a set $S \in \mathbf{P}(\mathbf{N})$ such that there is no $A \in \mathbf{P}(\mathbf{N})$ with $f(A) = S$.

Assume $S = \{2\}$. Suppose, for contradiction, that there exists A such that:

$$f(A) = A \cup \{1\} = \{2\}.$$

This implies $1 \in \{2\}$, which is false.

Therefore, no such A exists, and f is not onto.

4. [10 pts] Let g be a total function from A to B and f be a total function from B to C .

(a) If $f \circ g$ is one-to-one, then is g one-to-one? Prove or give a counter-example.

Answer: Yes

Explanation:

Assume $f \circ g$ is one-to-one.

Suppose $g(x_1) = g(x_2)$ for some $x_1, x_2 \in A$.

Apply f to both sides:

$$f(g(x_1)) = f(g(x_2)) \Rightarrow (f \circ g)(x_1) = (f \circ g)(x_2)$$

Since $f \circ g$ is one-to-one, it follows that $x_1 = x_2$.

Therefore, g is one-to-one.

(b) If $f \circ g$ is onto, then is g onto? Prove or give a counterexample.

Answer: No

Counterexample:

Let: $A = \{1, 2\}$, $B = \{a, b\}$, $C = \{x\}$

Define the functions:

$g : A \rightarrow B$ by $g(1) = a$, $g(2) = a$ (so g is not onto)

$f : B \rightarrow C$ by $f(a) = x$, $f(b) = x$ (so f is onto)

Now consider:

$$(f \circ g)(1) = f(g(1)) = f(a) = x$$

$$(f \circ g)(2) = f(g(2)) = f(a) = x$$

So the range of $f \circ g$ is $\{x\} = C$, meaning $f \circ g$ is onto.

Therefore, g is not onto

Relations

5. [10 Pts] For each of these relations decide whether it is reflexive, anti-reflexive, symmetric, anti-symmetric and transitive. **You need to give a proof or counter-example or a suitable reason for each property.** Justify your answers. R_1 and R_2 are over the set of real numbers.

(a) $(x, y) \in R_1$ if and only if $xy \geq 2$

Reflexive:

The relation is reflexive if $(x, x) \in R_1$ for all $x \in \mathbb{R}$. This would mean $x^2 \geq 2$ for all real numbers.

However, this is false.

Counterexample: If $x = 1$, then $x^2 = 1 < 2$, so $(1, 1) \notin R_1$.

R_1 is not reflexive.

Anti-reflexive

The relation is anti-reflexive if $(x, x) \notin R_1$ for all $x \in \mathbb{R}$. This would mean $x^2 < 2$ for all x , which is not true.

Counterexample: $x = 2 \Rightarrow x^2 = 4 \geq 2 \Rightarrow (2, 2) \in R_1$.

R_1 is not anti-reflexive.

Symmetric

The relation is symmetric if $(x, y) \in R_1 \Rightarrow (y, x) \in R_1$. Since $xy = yx$, I have:

$$(x, y) \in R_1 \iff xy \geq 2 \iff yx \geq 2 \iff (y, x) \in R_1$$

R_1 is symmetric.

Anti-symmetric

The relation is anti-symmetric if $(x, y) \in R_1$ and $(y, x) \in R_1 \Rightarrow x = y$.

Counterexample:

Let $x = 1, y = 2$.

$xy = 2 \geq 2$, so $(1, 2) \in R_1$ and $(2, 1) \in R_1$, but $1 \neq 2$.

R_1 is not anti-symmetric.

Transitive

The relation is transitive if $(x, y) \in R_1$ and $(y, z) \in R_1 \Rightarrow (x, z) \in R_1$, i.e., if $xy \geq 2$ and $yz \geq 2 \Rightarrow xz \geq 2$.

Counterexample: Let $x = 1, y = 2$, and $z = 1$.

$xy = 2 \Rightarrow (1, 2) \in R_1, \quad yz = 2 \Rightarrow (2, 1) \in R_1$

but $xz = 1 \times 1 = 1 < 2 \Rightarrow (1, 1) \notin R_1$

R_1 is not transitive.

(b) $(x, y) \in R_2$ if and only if $x = 4y$

Reflexive:

The relation is reflexive if $x = 4x$, or $3x = 0$, which only holds when $x = 0$.

Counterexample:

$$x = 1 \Rightarrow 1 \neq 4 \times 1 \Rightarrow (1, 1) \notin R_2$$

R_2 is not reflexive.

Anti-reflexive:

The relation is anti-reflexive if $(x, x) \notin R_2$ for all $x \in \mathbb{R}$.

Counterexample:

$$(0, 0) \in R_2, \text{ since } 0 = 4 \times 0$$

Conclusion R_2 is not anti-reflexive.

Symmetric:

The relation is symmetric if $(x, y) \in R_2 \Rightarrow (y, x) \in R_2$. Suppose $(x, y) \in R_2$, so $x = 4y$. This would require $y = 4x$, which is generally not true.

Counterexample:

$$x = 4, y = 1 \Rightarrow x = 4y \Rightarrow (4, 1) \in R_2, \text{ but } (1, 4) \notin R_2 \text{ since } 1 \neq 4 \times 4$$

R_2 is not symmetric.

Anti-symmetric:

A relation is anti-symmetric if $(x, y) \in R_2$ and $(y, x) \in R_2 \Rightarrow x = y$.

Suppose $x = 4y$ and $y = 4x$. Then:

$$x = 4y = 4(4x) = 16x \Rightarrow x = 0 \Rightarrow y = 0$$

So both x and y must be 0.

R_2 is anti-symmetric.

Transitive:

A relation is transitive if $(x, y) \in R_2$ and $(y, z) \in R_2 \Rightarrow (x, z) \in R_2$.

Suppose $x = 4y$, and $y = 4z \Rightarrow x = 4(4z) = 16z$

But for $(x, z) \in R_2$, $x = 4z$, not $x = 16z$, so this is generally false.

Counterexample: $(16, 4), (4, 1) \in R_2$, but $(16, 1) \notin R_2$ since $16 \neq 4 \times 1$

R_2 is not transitive.

6. [10 Pts] Consider relation R_3 defined below on the set of positive real numbers. Prove that R_3 is a partial order, *i.e.*, it is reflexive, anti-symmetric and transitive.

$$(x, y) \in R_3 \text{ if and only if } x/y \in \mathbb{Z}$$

Reflexive:

Show that for all $x \in \mathbb{R}^+$, $(x, x) \in R_3$.

$$\frac{x}{x} = 1 \in \mathbb{Z}$$

So $(x, x) \in R_3$ for all $x \in \mathbb{R}^+$.

Therefore, R_3 is reflexive.

Anti-symmetric:

Show that for all $x, y \in \mathbb{R}^+$, if $(x, y) \in R_3$ and $(y, x) \in R_3$, then $x = y$.

Assume: $\frac{x}{y} \in \mathbb{Z}$, $\frac{y}{x} \in \mathbb{Z}$

Let $\frac{x}{y} = m$ and $\frac{y}{x} = n$, where $m, n \in \mathbb{Z}^+$.

$$x = my \quad \text{and} \quad y = nx$$

Substitute $y = nx$ into $x = my$:

$$x = m(nx) = mn x \Rightarrow 1 = mn$$

Since m, n are positive integers and $mn = 1$, I can conclude $m = n = 1$.

Therefore:

$$\frac{x}{y} = 1 \Rightarrow x = y$$

Thus, R_3 is anti-symmetric.

Transitive:

Show that for all $x, y, z \in \mathbb{R}^+$, if $(x, y) \in R_3$ and $(y, z) \in R_3$, then $(x, z) \in R_3$.

Assume: $\frac{x}{y} \in \mathbb{Z}$, $\frac{y}{z} \in \mathbb{Z}$

$$\frac{x}{z} = \frac{x}{y} \cdot \frac{y}{z}$$

Since both $\frac{x}{y}$ and $\frac{y}{z}$ are integers, their product is also an integer: $\frac{x}{z} \in \mathbb{Z} \Rightarrow (x, z) \in R_3$

Thus, R_3 is transitive.

Since R_3 is reflexive, anti-symmetric, and transitive on $\mathbb{R}^+$, it is a partial order.

7. [10 Pts] Consider relation R_4 defined below on the set of real numbers. Prove that R_4 is an equivalence relation. What is the equivalence class of 2? of π ? Describe all the equivalence classes using set-builder notation.

$$(x, y) \in R_4 \text{ if and only if } x = y + n/2 \text{ for some integer } n.$$

Proving R_4 is an equivalence relation:

Reflexive:

For any $x \in \mathbb{R}$, let $n = 0 \in \mathbb{Z}$. Then

$$x = x + \frac{0}{2} \Rightarrow (x, x) \in R_4.$$

So R_4 is reflexive.

Symmetric:

Suppose $(x, y) \in R_4$, then

$$x = y + \frac{n}{2} \text{ for some } n \in \mathbb{Z} \Rightarrow y = x - \frac{n}{2} = x + \frac{-n}{2}.$$

Since $-n \in \mathbb{Z}$, it follows that $(y, x) \in R_4$.

R_4 is symmetric.

Transitive:

Suppose $(x, y) \in R_4$ and $(y, z) \in R_4$, then

$$x = y + \frac{n}{2} \quad \text{and} \quad y = z + \frac{m}{2} \quad \text{for some } n, m \in \mathbb{Z}.$$

Substituting, I get:

$$x = z + \frac{m+n}{2},$$

Since $m + n \in \mathbb{Z}$, it follows that $(x, z) \in R_4$.

R_4 is transitive.

Since R_4 is reflexive, symmetric, and transitive, it is an equivalence relation.

Equivalence classes:

The equivalence class of a real number a is:

$$[a] = \{x \in \mathbb{R} \mid (x, a) \in R_4\} = \left\{x \in \mathbb{R} \mid x = a + \frac{n}{2}, n \in \mathbb{Z}\right\}.$$

For $a = 2$:

$$[2] = \left\{x \in \mathbb{R} \mid x = 2 + \frac{n}{2}, n \in \mathbb{Z}\right\}.$$

For $a = \pi$:

$$[\pi] = \left\{x \in \mathbb{R} \mid x = \pi + \frac{n}{2}, n \in \mathbb{Z}\right\}.$$

All equivalence classes can be described as:

$$[a] = \left\{x \in \mathbb{R} \mid x = a + \frac{n}{2}, n \in \mathbb{Z}\right\}, \quad \text{for } a \in [0, 0.5).$$

8. [12 Pts] Let R_5 be the relation on $\mathbb{Z}^+ \times \mathbb{Z}^+$ where $((a, b), (c, d)) \in R_5$ if and only if $ad = bc$.

(a) Prove that R_5 is an equivalence relation.

Reflexive:

For any $(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+$,

$$ab = ba$$

so $((a, b), (a, b)) \in R_5$.

R_5 is reflexive.

Symmetric:

Suppose $((a, b), (c, d)) \in R_5$.

Then $ad = bc \Rightarrow cb = da$

so $((c, d), (a, b)) \in R_5$.

R_5 is symmetric.

Transitive:

Suppose $((a, b), (c, d)) \in R_5$ and $((c, d), (e, f)) \in R_5$.

Then:

$$ad = bc \quad \text{and} \quad cf = de$$

To show that $af = be$. I start by multiplying both equations:

$$(ad)(cf) = (bc)(de) \Rightarrow af \cdot dc = be \cdot dc \Rightarrow af = be$$

$((a, b), (e, f)) \in R_5$, so R_5 is transitive.

Since R_5 is reflexive, symmetric, and transitive, it is an equivalence relation.

(b) Define a function f such that $f(a, b) = f(c, d)$ if and only if $((a, b), (c, d)) \in R_5$.

Define function:

$$f(a, b) = \frac{a}{b}.$$

Then:

$$f(a, b) = f(c, d) \iff \frac{a}{b} = \frac{c}{d} \iff ad = bc \iff ((a, b), (c, d)) \in R_5$$

(c) Define the equivalence class containing $(1, 1)$.

Let $[(1, 1)]$ denote the equivalence class of $(1, 1)$.

Then:

$$[(1, 1)] = \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid a \cdot 1 = b \cdot 1\} = \{(a, a) \mid a \in \mathbb{Z}^+\}$$

(d) Describe the equivalence classes using set-builder notation.

Each equivalence class corresponds to a positive rational number $r \in \mathbb{Q}^+$. So, the equivalence class of (a, b) is:

$$[(a, b)] = \{(c, d) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid ad = bc\} = \{(c, d) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid \frac{c}{d} = \frac{a}{b}\}.$$

Thus, the set of all equivalence classes is:

$$\{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid \frac{a}{b} = r\}, \quad \text{for each } r \in \mathbb{Q}^+.$$

9. [20 Pts] Prove that these relations on the set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$ are equivalence relations.

(a) $R_6 = \{(f, g) \mid f(0) = g(0)\}$

Reflexive:

For any function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, I have $f(0) = f(0)$, so $(f, f) \in R_6$.

R_6 is reflexive.

Symmetry:

Suppose $(f, g) \in R_6$. Then by definition, $f(0) = g(0)$.

By the symmetry of equality, I have $g(0) = f(0)$, so $(g, f) \in R_6$.

R_6 is symmetric.

Transitive:

Suppose $(f, g) \in R_6$ and $(g, h) \in R_6$. Then $f(0) = g(0)$ and $g(0) = h(0)$.

By the transitivity of equality, $f(0) = h(0)$, so $(f, h) \in R_6$.

R_6 is transitive.

Since R_6 is reflexive, symmetric, and transitive, it is an equivalence relation on the set of functions from $\mathbb{Z}$ to $\mathbb{Z}$.

(b) $R_7 = \{(f, g) \mid \exists C \in \mathbb{Z}, \forall x \in \mathbb{Z}, f(x) - g(x) = C\}$

Reflexive:

For any function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, I have

$$\forall x \in \mathbb{Z}, \quad f(x) - f(x) = 0.$$

Let $C = 0 \in \mathbb{Z}$.

$$(f, f) \in R_7.$$

R_7 is reflexive.

Symmetry:

Suppose $(f, g) \in R_7$.

Then there exists $C \in \mathbb{Z}$ such that

$$\forall x \in \mathbb{Z}, \quad f(x) - g(x) = C.$$

$$g(x) - f(x) = -C.$$

Since $-C \in \mathbb{Z}$, it follows that $(g, f) \in R_7$.

R_7 is symmetric.

Transitive:

Suppose $(f, g) \in R_7$ and $(g, h) \in R_7$.

Then there exist $C_1, C_2 \in \mathbb{Z}$ such that

$$\forall x \in \mathbb{Z}, \quad f(x) - g(x) = C_1 \quad \text{and} \quad g(x) - h(x) = C_2.$$

Adding these equations, I get

$$f(x) - h(x) = C_1 + C_2.$$

Since $C_1 + C_2 \in \mathbb{Z}$, it follows that $(f, h) \in R_7$.

R_7 is transitive.

Since R_7 is reflexive, symmetric, and transitive, it is an equivalence relation on the set of functions from $\mathbb{Z}$ to $\mathbb{Z}$.